2026 9478 H2 Physics Preliminary Examination

Paper 2 (Mark Scheme)

1 **(a)** product of force and perpendicular distance from line of action of force to axis / pivot B1

(b) **(i)** (use principle of moments by taking moment about A,)
either a.c.w. moment = $T \times AD \sin 47^o$ or $T \times AD \cos 43^o$
(where $AD = 3.2 \sin (180^o - 53^o - 47^o) / \sin 47^o$ from sine rule)
or a.c.w. moment = $T \times 3.2 \cos 10^o$ or $T \times 3.2 \sin 80^o$ C1

c.w. moment = $11 \times 9.81 \times 6.5 \sin 53^o + 9.0 \times 9.81 \times (6.5 / 2) \sin 53^o$
(where $\sin 53^o$ can be replaced by $\cos 37^o$) C1

so $T = 250$ N A0

(ii) [Method 1 – component of force]
vertical: $F_y = (11.0 + 9.0) \times 9.81 - 250 \sin (180^o - 53^o - 47^o) = 152.8$ N
horizontal: $F_x = 250 \sin 80^o = 246.2$ N C1

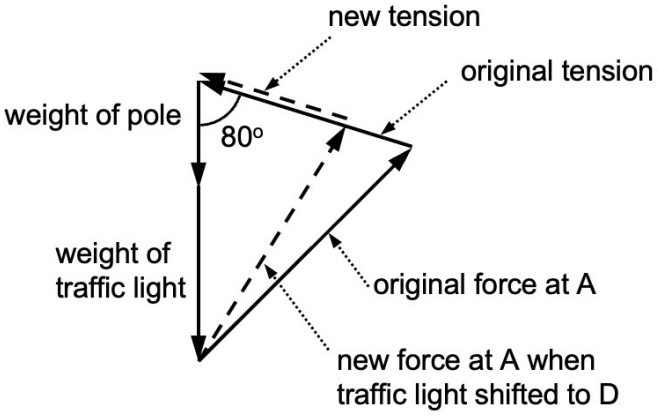
$F_A = (F_x^2 + F_y^2)^{1/2} = 290$ N A1

[Method 2 – cosine rule]
$F_A^2 = 250^2 + (20 \times 9.81)^2 - 2 \times 250 \times (20 \times 9.81) \cos 80^o$ (M1)

$F_A = 290$ N (A1)

(iii) moment of the weight of light about A decreases, so tension in cable decreases M1

(since tension remains in the direction of CD,)
either use cosine rule to show the force at A is smaller when tension decreases
or show with a well-labelled diagram that force A is smaller A1



2 **(a)** 4.0 N, downward (or statement must lead clearly) B1

show application and not definition of Newton's third law (e.g. candidate must state that the force exerted on ball A by ball B is equal and opposite to the force exerted on ball B by ball A) B1

(b) **(i)** not conserved B1

(ii) (average) resultant force acting on A (during contact)
= mg − contact force = $(0.50 \cdot 9.81) - 4.0 = 0.905$ N C1

change in momentum of A = $0.905 \cdot 0.50 = 0.4525$ N s C1

$0.50\ (v_f - 2.0) = 0.4525$ C1

$V_f = 2.9$ m s^{-1} A1

3 **(a)** $\lambda' = 191.9 \times 10^{-12} + \dfrac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(3.00 \times 10^{8})}(1 - \cos\cos\ 64.3°)$ M1

$\lambda' = 193.3 \times 10^{-12}$ m A0

(b) decrease in energy of photon ΔE
= $(6.63 \times 10^{-34})(3.00 \times 10^{8})(1 / 191.9 \times 10^{-12} - 1 / 193.3 \times 10^{-12})$ C1
= 7.5068×10^{-18} J

decrease in energy of photon transfer to kinetic store of electron or $\Delta E = P^2 / 2m$ B1

$P = [2\ (9.11 \times 10^{-31})\ (7.5068 \times 10^{-18})]^{1/2}$ M1

$= 3.698 \times 10^{-24}$ N s A0

(c) in direction perpendicular to incident photon,
momentum of deflected photon = $(6.63 \times 10^{-34} / 193.3 \times 10^{-12}) \sin 64.3°$ C1

momentum of deflected electron = $3.698 \times 10^{-24} \sin \alpha$ C1

clear statement stating conservation of momentum at right angle to the incident photon (e.g. incident photon has no momentum perpendicular to direction of incidence, so perpendicular component of electron's momentum equal and opposite to that of the deflected photon) M1

$3.698 \times 10^{-24} \sin \alpha = (6.63 \times 10^{-34} / 193.3 \times 10^{-12}) \sin 64.3°$

$\alpha = 56.7°$ or $56.69°$ A1

4 **(a)** period = $2\pi \times 6.7 \times 10^7 / 2.38 \times 10^4 = 17688$ s = 4.91 hours C1

period of particle <10 hours 14 minutes, so particle is not in krono-stationary orbit A1

(b) gravitational force on particle provides the centripetal force B1

$GMm / r^2 = mv^2 / r$
$M = v^2r / G = (2.38 \times 10^4)^2 \times 6.7 \times 10^7 / 6.67 \times 10^{-11}$ M1

$M = 5.7 \times 10^{26}$ kg A0

(c) $GM_T / x^2 = GM_S / (1.2 \times 10^9 - x)^2$

gravitational field at point due to Saturn = $GM_S / (1.2 \times 10^9 - x)^2$ C1

resultant field is zero, so $1.4 \times 10^{23} / x^2 = 5.7 \times 10^{26} / (1.2 \times 10^9 - x)^2$ C1

$x = 1.9 \times 10^7$ m A1

5 **(a)** Any three of the following: B3

- wave profile of stationary waves does not advance
- amplitude varies from zero at node to maximum at antinode.
- particles within the **internodal segment** oscillate in phase while particles in **adjacent internodal segments** oscillate anti-phase
- particles oscillate in SHM with same frequency except at the nodes
- no wavelength
- distance between adjacent nodes (or antinodes) equal half the wavelength of the waves that form the stationary wave

(b) waves from source reflected by reflective casing B1

incident and reflected wave (with same amplitude) overlap and superpose B1

(c) wavelength = $5.8 \times 2 = 11.6$ cm C1

speed = $2.45 \times 10^9 \times 0.116 = 2.8 \times 10^8$ or 2.84×10^8 m s^{-1} A1

(d) wavelength of microwave (11.6 cm) > 2 mm, wavelength of visible light (~10^{-7} m) < 2 mm B1

6 **(a)** total resistance of circuit decreases when resistance of T decreases B1

current in cell increases B1

terminal p.d. decreases (and resistance of P is constant) so current in R decreases B1

(b) p.d. of r = 1.50 – (0.200 × 6.00) = 0.30 V C1

current in cell = 0.30 / 0.12 = 2.5 A C1

resistance of thermistor = (0.200 × 6.00) / (2.5 – 0.200) = 0.52 Ω or 0.522 Ω A1

(c) $I = nAvq = (N / AL)\, Avq = (N / L)\, vq$
$v = IL / Nq = (0.200 \times 0.223) / (1.10 \times 10^{20} \times 1.60 \times 10^{-19})$ C1

$v = 2.53 \times 10^{-3}$ m s^{-1} A1

7 **(a)** **(i)** mass defect = [92 × 1.007 276 + (235 – 92) × 1.008665 – 235.043930] u M1
= 1.864 557 u

binding energy = 1.864 557 × 1.66 × 10^{-27} × (3.00 × 10^{8})^2 / 1.60 × 10^{-19} M1
= 1.741,030 · 10^{9} eV = 1741.030 MeV

binding energy per nucleon = 1741.030 / 235 M1

= 7.409 MeV A0

(ii) less (than) B1

(b) energy = (95 × 8.443) + (139 × 8.189) – (235 × 7.409) M1

= 199 MeV or 199.5 MeV A1

(c) number of reaction = 1.2 × 10^{-7} × 6.02 × 10^{23} C1
= 7.224 × 10^{16}

power = 7.224 × 10^{16} × 199.5 × 10^{6} × 1.60 × 10^{-19} / 25 × 10^{-3} M1

= 9.2 to 9.22 × 10^{7} W A1

8 **(a)** Force on sail ¿ $(\Delta v)\left(\frac{Nm}{t}\right)$ or ¿ $\left(\frac{N}{t}\right)\Delta p$ M1

$\Delta p \left(= 2mc = 2\frac{h}{\square}\right) =$ ¿ 2 × 6.63 × 10^{-34} / 1.0 × 10^{-6} C1

$\frac{N}{t} = \frac{P}{hf} = \frac{P}{hc} \times \lambda$ = (100 × 10^{9} × 1.0 × 10^{-6}) / (6.63 × 10^{-34} × 3 × 10^{8}) C1

F ¿ (2 × 6.63 × 10^{-34} / 1.0 × 10^{-6}) × (100 × 10^{9} × 1.0 × 10^{-6}) / (6.63 × 10^{-34} × 3 × 10^{8})

F ¿ 666.7 = 670 N A0

Award one mark if candidate uses $F = 2P / c$.

(b) **(i)** acceleration = 670 / 4.0 × 10^{-3} = 1.7 × 10^{5} m s^{-2} (accept 1.67 to 1.7) A1

(ii) $t = (v - u) / a$ = 0.20 × 3 × 10^{8} / 1.675 × 10^{5} = 358 s (358 – 360 s) A1

(c) **(i)** $\theta = s / r$ = 5.0 / 1.0 × 10^{10} M1

θ = 5.0 × 10^{-10} rad A0

(ii) laser that is not incident on the sail is wasted power B1

(iii) either sin (5.0 × 10^{-10} / 2) = $\lambda / b \Rightarrow b$ = 1.0 × 10^{-6} / sin (5.0 × 10^{-10} / 2) = 4000 m

or (for small angle) $b = \lambda / \frac{1}{2}\theta$ = 1.0 × 10^{-6} / 2.5 × 10^{-10} = 4000 m A1

(d) power absorbed = 0.001 × 100 × 10^{9} = 100 × 10^{6} W M1

thermal equilibrium: power absorbed = power radiated = $\varepsilon\sigma A T^{4}$ = 100 × 10^{6} M1

T = [100 × 10^{6} / (0.80 × $5.67 \times 10^{-8} \times 2 \times \pi \times 2.5^{2}$)]$^{1/4}$ = 2740 K or 2700 K A1

(e) low particle density / vacuum in space, hence sail cannot lose heat through conduction and convection B1

(f) **(i)** Only a fraction (or part) of the sail's surface is illuminated (by the laser) B1

(ii) Adjust beam width of the laser by shutting down some of the laser emitters B1

reduce the effective size of the array (hence changing the angle of diffraction of the beam or beamwidth) B1

(g) useful energy per decay = 0.067 × 5.48 × 10^{6} × 1.60 × 10^{-19} = 5.875 × 10^{-14} J C1

activity needed after 30 years A = 5.0 × 10^{-3} / 5.875 × 10^{-14} = 8.511 × 10^{10} s^{-1}

number of nuclei after 30 years $N = A / \lambda$ = 8.511 × 10^{10} / (ln 2 / 2.72 × 10^{9}) = 3.340 × 10^{20} C1

number of nuclei initially $N_0 = N / e^{-\lambda t}$ = 3.340 × 10^{20} / exp (–ln2 × 30 / 86.4) = 4.249 × 10^{20} C1

initial mass of plutonium = N_0 × mass of nuclei = $4.249 \times 10^{20} \times 238 \times 1.66 \times 10^{-27}$ M1

$= 1.7 \times 10^{-4}$ kg = 170 or 168 mg A1